

CHIEVRES AB, Belgium

WMO: 064320

Lat: 50.572N

Lon: 3.831E

Elev: 59

StdP: 100.62

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-4.0	-9.7	1.6	-5.6	-7.0	2.1	-3.2	13.9	8.0	12.5	8.3	3.0	40	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	29.7	20.1	27.5	19.4	25.4	18.3	21.1	27.5	20.1	26.0	19.2	24.2	3.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	13.8	23.6	18.0	13.1	22.3	17.2	12.4	21.4	61.2	27.6	57.7	26.1	54.6	24.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.4	8.2	DB	-8.4	33.0	3.7	1.7	-11.0	34.2	-13.2	35.2	-15.3	36.2	-17.9	37.4	(4)
(5)			WB	-8.8	23.1	3.5	1.2	-11.3	24.0	-13.3	24.7	-15.2	25.4	-17.8	26.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	3.5	4.4	6.9	9.6	13.0	16.2	18.2	18.0	14.9	11.7	7.3	4.4	(6)	
	DBStd	6.24	4.19	3.81	3.31	3.46	3.59	3.15	3.05	2.82	2.75	3.43	3.62	4.03	(7)	
	HDD10.0	821	203	158	104	49	10	0	0	0	1	22	96	178	(8)	
	HDD18.3	2881	460	391	354	263	169	80	41	41	108	207	332	433	(9)	
	CDD10.0	1076	1	1	8	36	104	186	254	248	148	74	13	3	(10)	
	CDD18.3	95	0	0	0	0	5	16	37	31	6	1	0	0	(11)	
	CDH23.3	952	0	0	0	9	59	174	351	302	53	3	0	0	(12)	
	CDH26.7	266	0	0	0	1	12	45	107	93	8	0	0	0	(13)	
(14) Wind	WSAvg	3.7	4.5	4.4	4.2	3.6	3.4	3.2	3.0	3.0	3.1	3.4	3.9	4.4	(14)	
(15) Precipitation	PrecAvg	767	61	50	68	51	63	73	68	57	57	73	76	70	(15)	
	PrecMax	1036	120	113	129	94	123	144	180	114	171	177	168	178	(16)	
	PrecMin	519	23	8	14	9	13	4	16	6	8	8	28	12	(17)	
	PrecStd	138	29	25	31	25	27	36	41	29	34	43	34	38	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	14.0	18.7	24.9	28.4	30.9	32.6	32.0	27.7	23.0	17.0	13.5	(19)	
		MCWB	11.1	9.0	11.9	16.0	18.8	20.4	20.7	20.8	19.2	17.3	14.2	11.6	(20)	
	2%	DB	11.7	11.8	16.0	21.6	24.8	27.7	29.6	29.3	24.8	20.2	14.9	12.3	(21)	
		MCWB	10.3	9.5	11.3	14.1	16.7	19.4	20.1	19.8	18.0	16.2	12.7	10.9	(22)	
	5%	DB	10.2	10.7	13.8	18.1	22.5	25.1	27.3	26.8	22.2	18.1	13.6	11.2	(23)	
		MCWB	8.7	8.9	10.2	12.5	15.8	18.4	19.5	19.1	16.9	15.0	11.9	9.9	(24)	
	10%	DB	9.0	9.6	12.1	15.7	19.9	22.7	25.0	24.2	20.2	16.7	12.4	10.0	(25)	
		MCWB	7.7	8.1	9.5	11.3	14.8	17.2	18.4	18.0	15.6	14.3	10.8	8.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	10.9	13.0	16.7	20.1	21.9	22.5	22.4	20.4	18.1	14.6	12.2	(27)	
		MCDB	12.5	12.5	16.5	23.4	26.5	27.9	29.0	29.7	26.2	21.7	16.3	13.0	(28)	
	2%	WB	10.4	9.9	11.8	14.6	18.1	20.2	21.2	20.7	18.9	16.6	13.0	11.1	(29)	
		MCDB	11.6	11.1	14.7	19.9	22.6	25.9	27.8	26.9	23.4	19.6	14.5	12.2	(30)	
	5%	WB	9.0	9.1	10.8	13.1	16.5	19.0	20.1	19.7	17.6	15.6	12.1	10.0	(31)	
		MCDB	10.1	10.4	13.4	17.4	21.0	24.1	26.0	25.4	20.7	17.8	13.5	11.1	(32)	
	10%	WB	7.8	8.2	9.7	11.8	15.2	17.7	19.1	18.6	16.5	14.6	11.0	8.8	(33)	
		MCDB	8.9	9.5	11.8	15.1	19.5	21.9	23.7	23.3	19.4	16.4	12.2	9.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.5	7.4	9.2	9.5	9.6	10.2	10.2	9.2	7.4	5.2	4.5	(35)	
		MCDBR	5.2	6.8	10.8	13.5	13.6	14.0	14.2	14.8	12.5	9.3	5.9	5.0	(36)	
	5% WB	MCWBR	4.6	5.2	6.9	7.4	6.9	7.2	6.4	6.6	6.4	5.8	4.6	4.4	(37)	
		MCWBR	5.4	6.0	9.1	12.0	11.8	12.2	12.7	13.3	10.5	8.2	5.5	4.8	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.328	0.349	0.385	0.417	0.428	0.424	0.426	0.417	0.389	0.366	0.340	0.325	(40)	
		taud	2.387	2.349	2.249	2.179	2.194	2.226	2.225	2.272	2.335	2.397	2.387	2.358	(41)	
	Ebn at Noon	Ebn at Noon	695	775	807	822	830	835	826	814	796	746	670	633	(42)	
		Edh at Noon	64	85	112	133	137	134	132	120	102	80	64	57	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.81	1.49	2.66	4.24	4.96	5.43	5.16	4.39	3.33	1.96	0.95	0.64	(44)	
	RadStd	RadStd	0.09	0.29	0.36	0.47	0.46	0.49	0.55	0.43	0.28	0.17	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page